

Assignment 1

Hardware 1 - Orientation to Health Technology

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The author confirms that the whole document is written without the use of any AI tools.

Summary

Scope:

The scope of the project is to make a working device that can read physiological information about a person via photoplethysmography, send the raw information to a cloud server for analysis, and then show the processed information to the user.

Application:

The device is intended to be a working proof of concept of a device which measures a person's stress and recovery. It will need to be able to perform three functions. Record new HRV analyses, read the results of previous HRV analyses, and show the history of previous HRV analysis results

Hardware device:

We will be using a crowtail pulse sensor, a raspberry pi pico w, an OLED display, a protoboard, and a rotary encoder to accomplish this project. Miscellaneous auxiliary components will likely also be used. The crowtail pulse sensor will be used to gather data from the user. The raspberry pi pico w will send the data to Kubios Cloud for analysis. The OLED display will show the user a graphical interface through which the device can be used. The protoboard will be used to connect many of the components together. And the rotary encoder will let the user interface with the device.

software development tools:

The device will be programmed in MicroPython. To more easily interface with the raspberry pi, we will use Thonny. Kubios Cloud will be used as the cloud server. And Wokwi may also be used for testing. MicroPython is used due to the raspberry pi pico w's limited memory (2 MB). Thonny is used to facilitate a significantly faster iterative process during coding. Kubios Cloud is used to analyze the data gathered by the pulse monitor and process it into informative and useful information we can show to the user.